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Police Officers' addiction frameworks and policy attitudes

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ABSTRACT

Police officers have frequent encounters with people who use drugs, either by making an arrest for a drug-related offense or responding to a drug overdose call. Yet, little is known about how police officers view drug addiction – as a disease, a moral failure, or something else – and how their frameworks for conceptualizing addiction impact their attitudes toward drug policies, including the use of naloxone. This research examined police officers' adherence to a moralistic addiction framework in relation to their support for treatment-oriented drug policies. Officers (N = 618) were surveyed about their beliefs on drug policy and the extent to which drug addiction was a product of one's morals or related to social or biological reasons. Results found that approximately 22% of the variance in drug policy attitudes could be explained by addiction frameworks and control variables. Officers who embraced a biological perspective of addiction were more supportive of policies that expanded treatment, including access to naloxone, and less punitive sanctions. Those with stronger moralistic views were less supportive of expanding treatment initiatives and endorsed expanding punitive sanctions. Officer age and education was positively related with expanding treatment and naloxone use while exposure to overdoses was negatively related to policy support. These results demonstrate that officers' frameworks about drug addiction play an important role in drug policy attitudes and, by extension, how they might interact with people who use drugs.

1. Introduction

Police officers often respond to drug overdose calls and administer naloxone, the overdose-reversing medication (Lurigio, Andrus, & Scott, 2018). In an increasing number of jurisdictions, police officers also participate in diversion programs where they help overdose survivors access drug treatment services (Formica et al., 2018; Knopf, 2017). Despite their interaction with people who use drugs, little is known about how police officers view drug addiction – as a disease, a moral failure, or something else – and how their addiction-conceptualizing frameworks impact their perspectives on various drug policies, including the use of naloxone.

1.1. Addiction frameworks and stigma

The brain disease model of addiction has become the prevailing theory of addiction in the United States, as shown by its wide support within federal agencies such as the National Institute on Drug Abuse (Volkow & Koob, 2015). This model argues that addiction is a chronic disease, resulting in physical brain changes where the afflicted becomes incapable of controlling drug use (Leshner, 2001). National studies in both the United States and Australia find that the majority of those surveyed agree that addiction is a disease (Lawrence, Rasinski, Yoon, & Curlin, 2013; Lock, Timberlake, & Rasinski, 2002; Meurk, Carter, Partridge, Lucke, & Hall, 2014). Studies with smaller and more specific samples have found similar support for the disease framework of addiction; many cite biological influences as more important than moralistic factors for understanding its causes (Alcoholism and Drug Abuse Weekly, 2010; Blendon & Benson, 2018; Furnham & Thomson, 1996; Jones, 2006; Murphy, 2017; Steenbergh, Runyan, Daugherty, & Winger, 2012). The disease model of addiction has also led to the expansion of programs that divert drug offenders into treatment rather than incarceration (Murphy, 2015).

Despite widespread acceptance that drug addiction is a disease, many addiction researchers object to this conceptualization and instead primarily view addiction as a choice or a learned condition related to one's environment (Baumeister & Vonasch, 2015; Foddy & Savulescu, 2010; Heyman, 2009, 2013; Lewis, 2017; Wilbanks, 1989). Researchers also critique the brain disease model of addiction for being overly reductionist and ignoring the multiple social, biological, and

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psychological processes related to addiction (Hall, Carter, & Forlini, 2015; Hammer et al., 2013; Wiens & Walker, 2015). These different models for understanding addiction may also be region-specific. For example, research on drug treatment providers showed that therapists in the United States were more likely to see addiction as a disease while providers in the United Kingdom were more likely to view addiction as a choice (Russell, Davies, & Hunter, 2011). More recent research concurs that treatment providers in the United States are more likely to endorse a disease model than their British, Australian, or Norweigian counterparts (Barnett, O'Brien, Hall, & Carter, 2020; Vederhus, Clausen, & Humphreys, 2016).

These studies also found that older treatment providers were more likely to endorse the disease view of addiction. Other research on both adolescents and adults has found a higher acceptance of the disease framework among older respondents and more negative attitudes toward people with addiction among those younger (Adlaf, Hamilton, Wu, & Noh, 2009; Barnett, Hall, Fry, Dilkes-Frayne, & Carter, 2018; Lytle, Apriceno, & Kowal, 2020; Meurk et al., 2014). Broadus and Evans (2015) also found that younger respondents were more likely to endorse a moral view of addiction. While age is typically associated with more conservative beliefs, this research suggests that knowledge and experiences of addiction might increase with age and result in less moralizing attitudes.

Another lingering view of addiction is the moral framework, which blames the individual for their substance use problem, due to having poor values, low morals, character flaws, immaturity, or lacking motivation (Broadus & Evans, 2015; Furnham & Thomson, 1996; Wilbanks, 1989). This moral view is thought to lead to the stigmatization of addiction. Proponents of the brain disease model argue that adopting a disease framework would lead to stigma reduction in the general population (Volkow, 2020). There is some support for this connection. In a recent study, students who believed addiction was a disease were less stigmatizing (as measured by blame, perceived dangerousness, and social distance) toward people who used heroin or other opioids than students who were more fatalistic about addiction (Kruis & Choi, 2020). Avery, Avery, Mouallem, Demner, and Cooper (2020) also found that the physicians and attorneys they surveyed who did not fully agree with the brain disease model showed more negative attitudes toward people with a substance use disorder.

The relationship between addiction views and stigma, however, is complex. In the Avery et al. (2020) study, respondents could choose only one model of addiction. Other research shows that a respondent's addiction framework can overlap moral, biological, and social explanations (Barnett et al., 2018; Broadus & Evans, 2015; Heather, 2017; Murphy, 2017; Ormston, Bradshaw, & Anderson, 2010). In short, one may view addiction as a disease but still also hold moralizing views by blaming the individual for either having the disease or for not getting better. As an example, a survey of Australians in 2012 showed that belief in addiction as a brain disease was not associated with reduced stigma toward people with alcohol or heroin use disorders, as measured by the Attitudes toward Mental Illness Questionnaire (Meurk et al., 2014). That same survey also showed that while the majority agreed that heroin addiction was a "disease," more than one third of respondents also agreed that "bad character" was a cause of heroin addiction (Meurk et al., 2014). Heather (2017) argues that the brain disease model is not the only way to reduce stigma; the public could be better educated about choice and learning models of addiction to reduce stigma.

Having contact with others who have used drugs and/or experienced addiction relates to addiction frameworks and stigma (Palamar, Kiang, & Halkitis, 2011). Those who had a history of addiction treatment were more likely to support a disease model of addiction while those with no personal experience of drug use were more likely to hold moral views of addiction (Broadus & Evans, 2015; Furnham & Thomson, 1996). Lytle et al. (2020) found that having friends or family members who experience addiction was significantly related to lower moralizing attitudes about drug use (i.e., viewing people with addiction as not trustworthy and having character flaws). Janulis, Ferrari, and Fowler (2013) also found that those with more familiarity of drug addiction, including having friends or families experiencing addiction, predicted lower levels of stigma toward people who use drugs.

Education could also relate to attitudes toward addiction and stigma. Corrigan and Watson (2007) found through a nationally representative sample that those with higher education in the United States were less likely to hold stigmatizing attitudes (perceived dangerousness, blame, and distance) toward people with psychiatric disorders. Similarly, Meurk et al. (2014) found that Australians with higher education levels revealed lower stigmatizing attitudes toward addiction.

1.2. Addiction frameworks and drug policies

How one views drug addiction could also impact policy decisions. For example, Lee and Rasinski (2006) found a positive correlation between viewing people with an addiction as blameworthy and lacking morality with an endorsement of greater punishment for first time possession of cocaine. Kennedy-Hendricks et al. (2017) showed that higher stigma toward people with a prescription opioid use disorder, as measured by perceived dangerousness, blame, and discrimination, led to greater support for punitive policies and less support for treatment-oriented responses. In their study, stigma was a stronger predictor toward policy attitudes than political ideology. Similarly, Murphy (2017) found that voters in California who held a stronger moralistic framework for conceptualizing addiction were less likely to support a ballot initiative that would expand treatment for drug offenders. Other research concurs that those who endorse a biological or social framework for understanding addiction tend to support drug policies that emphasize treatment over punishment (Giordano, 2014; Lancaster, Ritter, & Matthew-Simmons, 2013). Those who perceive drug users as diseased are also more likely to support harm reduction policies like heroin-assisted treatment and supervised injection facilities (Cruz, Patra, Fischer, Rehm, & Kalousek, 2007).

More research is needed to understand the relationship between addiction frameworks, stigma, and drug policy attitudes. It is also possible that addiction frameworks impact attitudes differently across groups. The current study attempts to better understand the relationship between addiction frameworks and policy attitudes among police officers, due to their central role as first responders to opioid overdoses.

1.3. Police officer attitudes

Despite police officers being on the front lines of the current opioid crisis, little research has examined their conceptualization of drug addiction and attitudes toward drug-related policies. The few studies that have examined police officers show that they often hold moralizing views toward people who use drugs as well as pessimism about the use of naloxone (Gnann, 2019; Green et al., 2013; Murphy & Russell, 2020). For instance, in an evaluation study of naloxone training in San Diego, police officers had high levels of agreement with statements like "people who overdose are to blame for their own overdose" (Wagner, Bovet, Haynes, Joshua, & Davidson, 2016). Formica et al. (2018) found that police officer stigma (i.e., negative views of people who use drugs) was a barrier to implementing programs where law enforcement assists overdose survivors in accessing drug treatment. Stigma can lead to feelings of frustration, burnout, and pessimism among police officers, especially for those who experience more exposure to overdoses (Murphy & Russell, 2020; Saunders et al., 2019).

This study examines how police officers' addiction framework (that is, whether they conceptualize addiction as a biological, social issue, or a moral flaw) influences their attitudes toward drug policies. Reflecting on the research on this topic, to our knowledge there has been no research conducted on police officers' addiction frameworks. We recognize this research as exploratory in nature, but also draw on previous studies that have found participants embracing a moralistic framework were less

likely to endorse initiatives to expand drug treatment (Murphy, 2017) and those embracing a biological/disease model were more likely to support treatment-oriented drug policies (Giordano, 2014; Lancaster et al., 2013). Further, additional research is needed to tease out the extent to which moralistic, biological, and social conceptualizations of addiction are related to drug policy attitudes.

We investigate the following research questions: Do police officers who embrace a strong moral addiction framework demonstrate stronger punitive attitudes and negative support for treatment-oriented drug policies? Will police officers who hold stronger biological and social frameworks for addiction show greater support for policies focused on expanding drug treatment and naloxone use? To what extent do officer characteristics relate to policy attitudes? Will supportive policy attitudes increase with officer age or education? Are officers with family or friends with drug addiction more supportive of drug policies? Will officer response to overdose incidents, administration of naloxone, or working in counties with higher overdose rates demonstrate lower support for treatment-oriented policies?

2. Materials and methods

2.1. Procedure

An anonymous, online survey was developed and approved by the Institutional Review Board (IRB) and conducted in 2018–2019. Seven counties in Pennsylvania were targeted because of differential rates of overdose, as published by the 2016 State Coroner and Medical Examiner reports. One of the objectives of this research was to examine how officer attitudes associated with exposure of overdose deaths, so counties demonstrating low and high rates of variability in overdose deaths in Pennsylvania were included. The county with the lowest death rate had 13.0 while the highest was 43.6 per 100,000.

Upon IRB approval, police chiefs and District Attorneys in the seven counties were contacted via email or phone about the study and assistance was requested to forward the survey link to their officers.

2.2. Participants

Participants included 618 full and part-time police officers from 21 counties in Pennsylvania, with the majority (n = 523, 85%) from the 7 targeted counties and the remaining participants were officers from counties not targeted. The link was likely shared via email to officers in nearby counties through social networks and professional association. The overall response rate for this survey constituted about 19% of all police officers working in the targeted counties[1].

Participants consisted of 515 male (83.3%) and 29 female (4.7%) officers and 7 (1.3%) identifying as non-binary. Approximately 11% (n = 67) of officers did not provide demographic information. Officer age ranged from 22 to 64, with an average of 41.36 (SD = 10.06). The majority of officers were white (n = 501, 91.6%), followed by Latino/Hispanic (n = 22, 4.0), Black/African American (n = 8, 1.5%), Asian American (n = 2, 0.4%), and "other" (n = 14, 2.6%) (71 participants did not provide a response). Most officers had a Bachelor's degree (n = 260, 46.9%), while others had some college (n = 120, 21.7%), Associate's degree (n = 101, 18.2%), Master's Degree (n = 9.6%), High School (n = 16, 2.9%), or Doctoral degree (n = 4, 0.7%). Almost 70% (n = 382, 68.8%) had drug addiction training in the academy and 80% (n = 443) reported addiction training in their current job[2]. Less than 3% (2.7%, n = 15) reported ever having a problem with their own drug/alcohol use, 327 (52.9%) stated they had a close family member who had problems with drugs/alcohol, and 309 (49.4%) had a close friend that has had problems with a drug/alcohol, and (n = 63 (10.2%) did not respond.

2.3. Measures

2.3.1. Dependent Variable: Policy attitudes

Six items were used to assess attitudes toward drug-related policies. These items have been used in previous research on drug policy attitudes, including studies of police officers' attitudes toward naloxone use. Items were rated on a scale where 1 = strongly disagree to 4 = strongly agree. Items included "Tax dollars should pay for drug treatment programs."; "Increasing access and utilization of Narcan/Naloxone is a good solution to the current opioid problem."; and 4 items that were reverse coded; "Increasing access and utilization of Naloxone provides individuals with a substance use disorder an excuse to continue their drug use"; "Criminal penalties should be increased for drug possession." "Criminal penalties should be increased for selling drugs."; and "There should be a limit on how often someone who overdoses be administered Narcan." Items were averaged and higher scores indicate less punitiveness and greater support of treatment-oriented policies, Cronbach's α = 0.77.

2.3.2. Independent Variables: Addiction Frameworks.

Participants rated a list of eight items designed to reflect either a moral, biological or social framework of drug addiction by indicating the extent to which participants believed each item was a major, minor, or not a factor in drug addiction. Moralistic attitudes comprised of two items, "lacking willpower" and "lacking good moral values". Responses were scored as 2 = "a major factor", 1 = minor factor, and 0 = not a factor. All scores were reverse coded to reveal higher scores indicating stronger support for that addiction framework.

Biological items included three items: "Being born with a genetic trait that makes a person more likely to become an addict"; "Having a psychological illness such as depression or anxiety" and "Being prescribed opioids by a doctor for legitimate medical reasons."

Lastly, social framework included 3 items that focused on sociological explanations for addiction. They included, "Living in a situation where there are social pressures to take drugs"; "Living in a situation where other family members use drugs"; and "Having major personal family, financial or job problems."

2.4. Control variables

2.4.1. Age, Education, years of Experience, and friends and family with addiction[3].

We examined possible control variables such as participant age, education, number of deaths per county, friends or close family members who have had drug problems, how many times in the past 12 months they responded to a call about someone overdosing on drugs and how many times in the past 12 months they administered an overdose reversing drug such as Narcan to someone who had overdosed. Participant education was measured as 1 = High school diploma, 2 = Some college, 3 = associates degree, 4 = Bachelor's degree, 5 = Master's degree, and 6 = Doctoral degree. Officers were also asked 2 questions (yes/no) designed to identify whether they had close family members or close friends who have had problems with drugs or alcohol. Officers reported the number of years they served as a police officer. Responses

[1] This percentage was calculated by examining online reports of the number of police officers in the targeted counties and by contacting local police departments directly.

[2] This question did not specify the type of training received so it is possible that officers were including naloxone training as training related to drug addiction.

[3] We were intending to use officer's race as a control variable, given its importance in understanding attitudes toward drug policies and addiction. However, our sample was overwhelmingly white (92%), reflecting the police officer demographics of the areas we surveyed, and did not yield a large enough sample of black or Latinx officers for comparison.

ranged from 1 = less than one year, 2 = 1 year, 3 = 2 years, 4 = 3 years, 5 = 4 years, 6 = 5 years and 7 = more than 5 years. Table 1 displays the mean number scores for all control variables.

2.4.2. Experience with overdose calls & number of deaths per county

Two items measured the officer's experience with overdose calls and administering Naloxone; "How many times in the past 12 months have you responded to an overdose-related call", and "How many times in the past 12 months have you administered the overdose reversing drug Naloxone/Narcan to someone who overdosed" with responses ranging from "Never" to "More than 5 times." Lastly, number of overdose deaths per 100,000 people in each county was included as a control variable.

3. Results

Preliminary analyses using Pearson Product Moment Correlations with the dependent variable of interest found significant positive relationships with participant age $r(493) = 0.19$, $p < .001$ and level of education, $r(544) = 0.09$, $p < .03$. The two items assessing number of times officers responded to an overdose, ($r(556) = -0.20$, $p < .0001$), and administered Narcan, ($r(557) = -0.12$, $p < .01$) were inversely related to policy support. Officer experience, number of deaths per county and having a close friend or family member with a substance abuse problem were not significantly related to the dependent variable and removed from further analyses. Officer age, education and response to overdose and administering Narcan remained in analyses as control variables.

Correlations were then conducted to examine addiction framework items and attitudes toward drug policies (see Table 1). Table 1 shows significant relationships among demographic variables and the dependent variable. Results revealed that the morality framework variables, including addiction is due to lack of willpower ($r(552) = -0.21$, $p < .01$) and lack of good morals $r(553) = -0.30$, $p < .01$), were negatively related to support for treatment-oriented drug policies. In contrast, genetics, psychological illness, and being prescribed opiates (biological factors) and all social factors were positively related to treatment-oriented drug policies. Significant negative relationships were found for number of times an officer responded to an overdose ($r(556) = -0.20$, $p < .01$) and number of times administering naloxone and policy support ($r(557) = -0.12$, $p < .01$), indicating the more often officers respond to overdose incidents and administer Narcan, the less likely they were to endorse treatment-oriented drug policies. Significant and positive relationships were also found for education ($r(544) = 0.09$, $p < .05$) and officer age ($r(493) = 0.19$, $p < .01$), indicating as age and education increase so does treatment-oriented policy support.

To evaluate our research questions and investigate how well measures of morality, biology, and social frameworks predict drug policy support, controlling for officer age, education, and response to overdose situations, we ordered the sets of predictors such that control variables were entered into the first step of the equation, followed by predictors of morality, biology, and social. Therefore, one multiple regression with 4 steps was conducted using the enter method to predict the criterion variable officer support of treatment-oriented drug policies (Table 2).

The first set in the regression analysis included the officer characteristic variables age, education, and response to overdose incidents. Tests to determine whether the data met the assumption of collinearity indicated multicollinearity was not a concern. Tolerance scores in Model 1 ranged from 0.73 to 0.98 and VIF ranged from 1.01 to 1.36. The results of this analysis indicated that officer characteristics such as education and age were positively related to treatment-oriented policy attitudes while number of times responding to overdose incidents was negatively related to treatment-oriented policies, $R^2 = 0.071$, adjusted, $R^2 = 0.063$, *FChange* (4479) = 9.11, $p < .0001$.

The second step evaluated whether addiction conceptualizations that included lack of morals and willpower predicted policy attitudes over and above officer characteristics. Tolerance scores in Model 2 ranged

Table 1
Correlations of Policy Support with Independent Variables and Control Variables.

Variables	*n*	*M*	*SD*	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Policy Support	561	2.12	0.67	—												
2. Lack willpower	552	1.61	0.56	−0.21**	—											
3. Lack morals	553	1.46	0.64	−0.30**	0.41**	—										
4. Genetic	554	1.05	0.71	0.27**	−0.06	−0.09*	—									
5. Psych Illness	554	1.47	0.59	0.23**	0.05	0.02	0.39**	—								
6. Prescribed opiates	554	1.43	0.62	0.18**	-0.01	−0.10**	0.15**	0.20**	—							
7. Social pressure	554	1.63	0.52	0.13**	0.02	−0.03	0.18**	0.31**	0.18**	—						
8. Family members	554	1.68	0.52	0.15**	0.01	0.01	0.17**	0.22**	0.16**	39**	—					
9. Personal/finance	554	1.09	0.66	0.21**	−0.03	−0.04	0.26**	0.35**	0.22**	0.22**	0.28**	—				
10. # times respond	556	2.41	1.54	−0.20**	0.06	0.07	−0.06	−0.09*	−0.03	−0.07	−0.06	−0.05	—			
11. # times Narcan	556	1.83	1.23	−0.12**	0.04	0.02	−0.10*	−0.10*	−0.01	−0.04	−0.03	−0.06	0.48**	—		
12. Age	493	41.36	10.64	0.19**	−0.08	−0.08	0.09*	0.07	0.04	0.00	0.04	0.07	−0.20**	0.16**	—	
13. Education	544	3.41	1.04	0.09*	−0.05	−0.02	0.02	−0.00	0.00	0.02	0.00	−0.02	−0.06	−0.05	−0.09	—

Note: ** Correlation is significant at the 0.01 level (2-tailed); * Correlation is significant at the 0.05 level (2-tailed).

Table 2
Multiple Regression Analysis Moral/ Biological/ Social Variables.

Model	B	95% CI	β	t	p
Step 1 Officer Characteristics					
(Constant)	1.64	[1.26 2.01]		8.68	0.000
Education	0.064	[0.008 0.121]	0.099	2.23	0.026
Age	0.011	[0.005 017]	0.16	3.53	0.000
How many times respond	−0.065	[−11 -0.021]	−0.15	−2.89	0.004
How many times administer	−0.009	[−0.062 0.044]	−0.01	−0.33	0.74
Step 2 (Moral/Willpower)					
(Constant)	2.17	[1.76 2.58]		10.45	0.000
Education	0.057	[0.002 0.111]	0.08	2.05	0.041
Age	0.009	[0.004 0.015]	0.14	3.18	0.002
How many times respond	−0.05	[−0.094–0.008]	−0.11	−2.35	0.019
How many times administer	−0.01	[−0.064 0.037]	−0.02	−0.51	0.604
Lack Willpower	−0.07	[−0.178 0.040]	−0.06	−1.24	0.212
Lack Morals	−0.25	[−0.34–0.156]	−0.24	−5.22	0.000
Step 3 (Biological)					
(Constant)	1.79	[1.38 2.20]		8.60	0.000
Education	0.054	[0.002 0.106]	0.08	2.03	0.043
Age	0.008	[0.002 0.013]	0.11	2.80	0.005
How many times respond	−0.052	[−0.094 –0.011]	−0.12	−2.49	0.013
How many times administer	0.003	[−0.046 0.052]	0.00	0.118	0.90
Lack Willpower	−0.075	[−0.180 0.030]	−0.06	−1.45	0.145
Lack Morals	−0.233	[−0.324 –0.142]	−0.24	−5.32	0.000
Psychological Illness	0.179	[0.082 0.275]	0.16	3.63	0.000
Genetics	0.150	[0.067 0.233]	0.15	3.54	0.000
Step 4 (Social)					
(Constant)	1.51	[1.06 1.97]		6.61	0.000
Education	0.059	[0.007 0.111]	0.09	2.23	0.026
Age	0.008	[0.002 0.013]	0.11	2.73	0.007
How many times respond	−0.048	[−0.089 –0.007]	0.11	−2.29	0.022
How many times administer	0.001	[−0.047 0.050]	0.00	0.05	0.960
Lack Willpower	−0.077	[−0.182 0.027]	−0.06	−1.45	0.146
Lack Morals	−0.233	[−0.324 -0.142]	−0.22	−5.03	0.000
Psychological Illness	0.124	[0.021 0.226]	0.11	2.37	0.018
Genetics	0.126	[0.042 0.209]	0.13	2.95	0.003
Social Pressure	0.026	[−0.089 0.142]	0.02	0.45	0.652
Other Family Members	0.062	[−0.051 0.175]	0.04	1.07	0.284
Personal/Financial Problems	0.068	[−0.024 0.160]	0.04	1.45	0.145
Prescribed Opiates	0.093	[0.003 0.183]	0.08	2.02	0.043

from 0.72 to 0.97 and VIF ranged from 1.02 to 1.38. The two items accounted for a significant portion of the variance in officer policy attitudes after controlling for the effects of officer characteristics, $R^2 = 0.145$, adjusted, $R^2 = 0.134$, *FChange* (2, 477) = 20.72, $p < .0001$. Results found that lacking good morals was related to less support for treatment-oriented drug policies, accounting for approximately 7.5% of the variance. Lacking willpower was not a significant predictor for supporting treatment-oriented drug policies.

The third regression analysis investigated the extent to which biological frameworks such as genetics, psychological illness, and being prescribed opiates affected the criterion variable after controlling for moral frameworks and officer characteristics. Tolerance scores in Model 3 ranged from 0.72 to 0.97 and VIF ranged from 1.03 to 1.38. Results found these three items accounted for a significant proportion of variance in policy attitudes, $R^2 = 0.226$, adjusted, $R^2 = 0.211$, *FChange* (3, 474) = 16.49, $p < .0001$. Results revealed a positive relationship indicating belief that genetics, psychological illness, and being prescribed opiates was related to increased support for treatment-oriented policies and accounted for an additional 8% of the variance.

Lastly, a regression analysis was conducted to assess the role of social factors as they relate to the criterion variable policy attitudes while controlling for all other variables. Tolerance scores for Model 4 ranged from 0.71 to 0.97 and VIF ranged from 1.03 to 1.40, (Durbin-Watson value = 1.85). Results showed none of the social factors contributed significantly to the amount of variance to the equation, $R^2 = 0.23$, adjusted, $R^2 = 0.21$, *FChange* (3, 471) = 1.69, $p > .05$.

4. Discussion

As hypothesized, addiction frameworks were strongly related to attitudes about drug policy. Those with a stronger biological framework for understanding addiction were more likely to support policies that expand treatment, decrease punishment, and increase access to naloxone. Those with a stronger moralistic framework were significantly less likely to support such policies. It is not surprising that those with moralistic views of addiction would be less likely to support expanding treatment and naloxone use, given that previous research has found that those who hold moralistic views of people who use drugs tend to be more punitive and less supportive of policies that expand drug treatment (Giordano, 2014; Kulesza, Teachman, Werntz, Gasser, & Lindgren, 2015; Lee & Rasinski, 2006; Luty & Grewal, 2002; Matheson et al., 2014; Murphy, 2017; Timberlake, Lock, & Rasinski, 2003). Still, it is noteworthy how strong the addiction frameworks related to attitudes. Even when adding significant control variables, such as officer age, education, and exposure to overdoses, belief that addiction is caused by lacking morals remained a strong predictor of drug policy attitudes. In future research on drug policy attitudes, researchers should consider how addiction frameworks could relate to attitudes and behaviors. While we operationalized "lacking willpower" as an indicator of a moral framework for addiction, the results showed that, once control variables were added, belief that addiction is caused by lacking willpower was no longer a significant predictor of policy attitudes. It is possible that the view that someone with addiction lacks willpower is not a purely moral one (Baumeister & Vonasch, 2015; Courtwright, 2010; Wilbanks, 1989). A contradictory model to the biological or disease model is one of free choice and decisions to take drugs (Baumeister & Vonasch, 2015; Heyman, 2009; Lewis, 2011). Models of addiction are still hotly debated among scholars and the issue of lacking free will plays a large role in this debate. While lacking free will was significantly related to lacking good morals in the current study, it failed to predict policy attitudes. Perhaps lacking free will and similar variables that assess agency, choice, and self-control should be examined as an explanatory construct for support in addiction policy decisions in future research.

The current research also showed that adherence to biological influences of addiction contributed 8% of the variance in policy support to expand treatment. This is consistent with previous research (Avery et al., 2020; Meurk et al., 2014) that demonstrated those who endorse the biological/disease model were less likely to stigmatize drug addiction. The current study expands upon previous research to reveal this framework as a contributor of drug policy support among police officers.

In addition to addiction frameworks, other variables were significantly related to policy attitudes. As expected, education was positively related to supporting treatment-oriented policies. Those with higher educational attainment were less punitive and in favor of expanding treatment and naloxone use. Older police officers, regardless of addiction framework, held more supportive views of policies designed to expand treatment and naloxone use. Previous research on stigma shows that older people tend to be less stigmatizing than younger people, but no research to our knowledge has examined this phenomenon among police officers. Perhaps older police officers are more supportive of treatment and naloxone use because of their experience as police officers during the era of mass incarceration for drug offenses. Such "war on drugs" policies demonstrated little public support years after they were enacted (Cook & Brownstein, 2019; Garland & Bumphus, 2012). While there is very little research on police officer attitudes toward drug

policies, one study found that police officers in one metropolitan department disagreed with many of the previous drug war strategies (Jorgensen, 2018). Older officers might also have been exposed to more initiatives that divert drug offenders into treatment and witnessed positive impacts of such policies (VanderWaal, Chriqui, Bishop, McBride, & Longshore, 2006).

How often an officer responded to overdose calls in the past year was also related to drug policy attitudes. As expected, officers who responded to more overdoses were less supportive of expanding treatment and naloxone use. This may be the result of the emotional exhaustion and compassion fatigue that could arise from responding to overdose calls, especially if the officer sees the same person multiple times (Green et al., 2013; Saunders et al., 2019). While the current research did not find administration of naloxone to be a predictor of policy attitudes, once other variables were added to the model, the results suggest that the mere exposure to overdoses, and not necessarily reviving a person who overdoses, is an important factor in understanding how officers view drug policies. Although previous research has shown exposure to addiction through family/friends to be related to attitudes toward people who use drugs, our findings show that these variables were not related to policy attitudes. It appears that, among police officers, their frequent exposure to people who use drugs as part of their job duties may negate the impact that having family or close friends with addiction has on attitudes.

The model with addiction frameworks and control variables accounted for 22% of the variance in policy attitudes, suggesting that there are other unmeasured factors related to attitudes. It was also somewhat surprising that none of the variables that endorse a more social framework for addiction were significantly related to policy attitudes. Future research should move beyond existing paradigms and include a more thorough examination of factors like empathy, self-control, and self-regulation and how they relate to addiction frameworks and attitudes (Baumeister & Vohs, 2003; Baumeister & Vonasch, 2015). Bivariate correlations also showed a significant negative relationship between administering naloxone and biological factors of addiction. Those who administered naloxone more frequently were less likely to view addiction as the result of biological factors. However, there was no significant correlation with support for a moral framework and naloxone administration, so it is not entirely clear how job duties, like administering naloxone, influence addiction frameworks and attitudes.

A significant limitation to the current study is the inability to control for officer race. Future research needs to incorporate the role of race in understanding officer addiction frameworks and policy attitudes. Our sample was overwhelmingly white due to the targeted counties sampled. Additional research is needed that includes officers of color so that their attitudes and frameworks can be assessed. Given that officers of color have shown to have more progressive views toward drug policies, including syringe exchange programs (Davis et al., 2014), than their white counterparts, it is crucial to incorporate the role of race in police officer attitudes. The sample also had a low percentage of female officers represented so it is difficult to assess the impact of officer sex on addiction frameworks and policy attitudes. While Pennsylvania does not publicize data about the number of female police officers in local police departments, females make up about 5% of state police officers. Future research will have to oversample female police officers to better examine their perspectives and experiences.

5. Conclusion

This study contributes to the scant literature on addiction frameworks and policy attitudes. First, it focuses on police officers, who are often on the front lines of the current opioid overdose crisis. Such research can assist in the implementation of training programs designed to assess police officers' perspectives about addiction and educate them about addiction processes, given their high exposure to people who use drugs and overdoses. Such training might minimize burnout, emotional exhaustion, and increase empathy toward people with substance use disorders. Second, this research illuminates the strong role that one's addiction framework has on their views of punishment, drug treatment, and naloxone policies. Having a moralistic addiction framework could lead to stigmatizing views about drug treatment and other policies. If officers see people who use drugs as morally inferior to the rest of the population, they may be less supportive of policies that could help people access treatment. As police departments adopt more practices to encourage people who use drugs into treatment, it is essential to reduce stigma among officers who may be responsible for such tasks.

Author disclosure

Authors Murphy and Russell designed the study and wrote the protocol. Author Murphy conducted literature searches and provided summaries of previous research studies. Author Russell conducted the statistical analysis. Author Murphy wrote the first draft of the manuscript and all authors contributed to and have approved the final manuscript.

CRediT authorship contribution statement

Jennifer Murphy: Project administration, Conceptualization, Methodology, Writing - original draft, Writing - review & editing. **Brenda Russell:** Formal analysis, Validation, Methodology, Writing - original draft, Writing - review & editing.

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